



Resonant tunnelling diodes based on twisted black phosphorus homostructures

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Atomically thin materials can be used to build novel forms of conventional semiconductor heterostructure devices. One such device is a resonant tunnelling diode, which can exhibit negative differential resistance and usually consists of a quantum-well structure between two barrier layers. Here, we show that a twisted black phosphorus homostructure can be used to create a resonant tunnelling diode. The devices have a trilayer structure in which a thin non-degenerate black phosphorus layer is sandwiched between two thicker degenerate black phosphorus layers. The interlayer coupling strength depends sensitively on the twist angle between the layers, and thus the twist angle can be used to control the vertical transport behaviour, from ohmic to tunnelling. Because resonant tunnelling through quantum-well states occurs without the need for a physical tunnelling barrier, our devices exhibit a higher tunnelling conductance and negative differential resistance peak-to-valley current ratio than resonant tunnelling diodes based on van der Waals heterostructures.

Resonant tunnelling diodes (RTDs) are quantum-well heterostructures that typically consist of a semiconductor layer between two physical barriers. Electrons or holes tunnelling through the discrete quantum-well states (QWSs) of the RTD results in non-monotonic current–voltage characteristics, such as negative differential resistance (NDR)^{1–3}. The devices are also capable of high-speed operation because tunnelling is, in principle, a very fast process. As a result, RTDs are of use in high-frequency devices such as oscillators⁴, mixers^{5,6} and high-speed switches¹. Van der Waals (vdW) heterostructures made from two-dimensional (2D) materials, including tunnel devices based on transition-metal dichalcogenides^{7–11} and hybrid tunnel devices based on black phosphorus (BP)¹², can also exhibit NDR current–voltage characteristics. The physical mechanisms of the NDR characteristics in these devices are typically the result of Esaki diodes⁹ and type-III broken-gap junctions^{13,14} (Supplementary Table 1). NDR has also been observed in graphene/boron nitride/graphene heterostructures^{15,16} where the effect is due to the resonant tunnelling of carriers, resulting from the matching of the electron energy momentum phase space at the two graphene electrodes. However, owing to the presence of a highly resistive physical barrier, RTDs based on vdW heterostructures currently show suppressed peak-to-valley current ratios (PVCR < 4)^{10,11,16,17} and limited charge current.

Highly anisotropic layered BP^{18–20} exhibits a direct and narrow bandgap of 0.39 eV with excellent transport properties (carrier mobilities of up to 10^3 – 10^4 cm² V^{−1} s^{−1})^{21,22}. The *sp*³ bonding between phosphorus atoms causes the puckered structure of BP (Fig. 1a) and the large electronic band anisotropy. The non-planar *sp*³ bonding is also responsible for the much larger interlayer coupling in BP compared to the planar *sp*² bonding in graphene^{23,24}. BP, as prepared, is generally a degenerate semiconductor for thicknesses above ~50 nm (refs. 14,25). For example, the carrier density increases from

2.1×10^{11} cm^{−2} to 7.5×10^{13} cm^{−2} when the BP thickness is increased from 4.5 nm to 90 nm, respectively (Supplementary Fig. 1).

In this Article, we show that a twisted trilayer structure that consists of a thin non-degenerate BP layer (7–14 atomic layers) sandwiched between two thicker (>50 nm) degenerate BP flakes (Fig. 1a,b and Supplementary Fig. 2) can exhibit NDR behaviour with a large tunnelling conductance and PVCR (up to 9 at 4.8 K). The orthogonally stacked homostructure exhibits weak interlayer coupling strength, and the electronic decoupling between the twisted BP layers—the amount of which depends on the twist angle—leads to well-resolved QWS within the sandwiched multilayer, as well as demonstrating an approach to creating RTDs where the NDR effect is controllable by the twist angle between layers of the same material. Resonant tunnelling occurs when the energy and momenta of the top and bottom BP match those of the QWS due to energy and momentum conservation. We examine these resonant states as a function of applied bias and thicknesses using electrical transport measurements.

BP trilayer homojunctions

We fabricated BP trilayer structures using a standard polydimethylsiloxane (PDMS) stamping technique by stacking 7–14-layer (L) BP multilayers between two degenerate BP flakes, both with thickness of >50 nm (Methods). These top and bottom degenerate BP flakes are denoted t-BP and b-BP, respectively. The middle-BP (m-BP) flake is twisted 90° with respect to the t-BP and b-BP flakes (Fig. 1a). Figure 1b presents cross-sectional high-resolution transmission electron microscopy (HRTEM) images of our measured devices based on a 90° twisted trilayer BP homostructure (trilayer orthogonal homojunction hereafter). We observed high-quality interfaces in our devices, and the different crystal orientations at t-BP/m-BP or b-BP/m-BP junctions could also be distinguished in scanning

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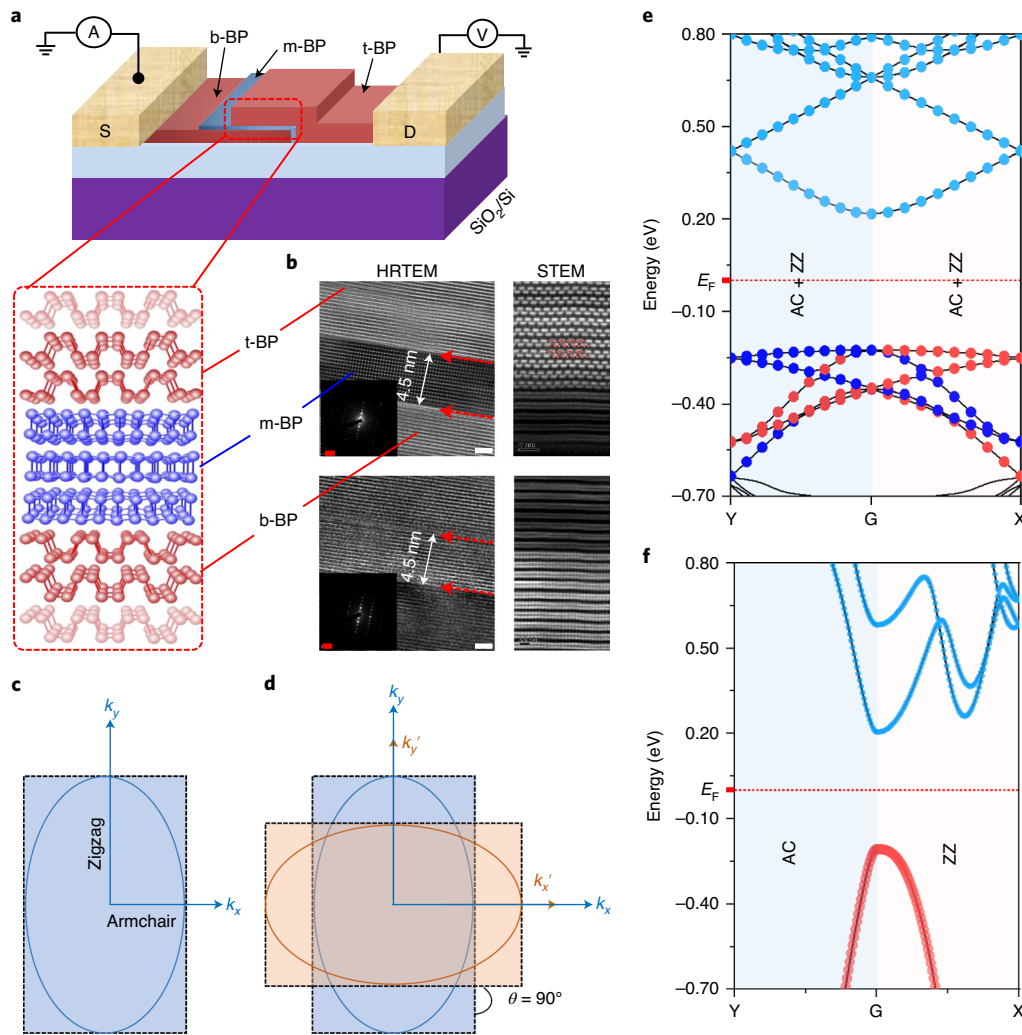


Fig. 1 | Device and band structures of BP homojunctions with twist angles of -90° and 0° . **a**, Schematic of a trilayer orthogonal homojunction device. The zoomed-in region shows that the middle-BP (m-BP) flake is twisted by 90° compared to the top-BP (t-BP) and bottom-BP (b-BP) flakes. **b**, HRTEM and STEM images of the trilayer orthogonal homojunction (top) and trilayer linear homojunction (bottom). Individual flakes are labelled. Selected area electron diffraction patterns taken from the t-BP region (insets). The STEM image was acquired near the t-BP/m-BP interface. The appearance of armchair (AC) and zigzag (ZZ) directions for the t-BP and m-BP, respectively, in the STEM image of one of the measured trilayer orthogonal homojunction devices is direct evidence of the 90° twisted BP structures. Dashed arrows in the HRTEM images indicate the junctions between flakes. Scale bars, 1 nm (**b**, main panel) and 2 nm^{-1} (insets). **c**, The Brillouin zone of monolayer BP. **d**, The Brillouin zone corresponding to 90° twisted BP bilayers. The elliptical Fermi surface has also been schematically illustrated within the first Brillouin zone (k_x, k_y and k'_x, k'_y are the momentum wave vectors). **e, f**, First-principles calculations for band structures of 90° twisted BP bilayers (**e**) and naturally (AB) stacked BP bilayer (**f**). The Fermi level (E_F) is shown as dotted horizontal lines. Different colours (blue and red) in the valence-band features in **e** indicate different BP layers. For the AB-stacked BP bilayer, ZZ and AC directions in real space are labelled, corresponding to the G-X and G-Y directions, respectively. However, 90° twisted BP layers exhibit isotropic features, that is, a combination of ZZ and AC directions along G-X and G-Y.

transmission electron microscopy (STEM) images, which clearly show the atomic armchair arrangement in t-BP and zigzag arrangement in m-BP (Fig. 1b). As a control experiment, we fabricated a trilayer BP homojunction where the t-BP, m-BP and b-BP were intentionally aligned linearly with a high degree of precision and with rotational stacking faults of only $2\text{--}5^\circ$, as demonstrated in the HRTEM and STEM images, which show similar atomic arrangements in the t-BP (b-BP) and m-BP layers (Fig. 1b). The crystal direction was also confirmed by calculating the lattice parameters from a selected area electron diffraction pattern acquired on a t-BP flake, as shown in the inset of Fig. 1b. The lattice constants were found to be 0.45 nm (armchair) and 0.32 nm (zigzag), respectively. These lattice parameters are in excellent agreement with previous reports^{26,27}.

The Brillouin zone of monolayer BP with its high symmetry points is shown in Fig. 1c. Owing to the band anisotropy of BP, the Fermi surface is described by an ellipse, with the long axis along the zigzag crystal direction. For the 90° twisted bilayer BP, the Fermi surfaces of the two BP flakes are misoriented, as depicted in Fig. 1d. The mismatch in the Fermi surfaces plays an important role in the vertical transport across the layers. We carried out first-principles calculations using the Perdew–Burke–Ernzerhof (PBE) functional (Methods) on bilayer BP with twist angles of 90° (Fig. 1e) and 0° (Fig. 1f). Clearly, the band dispersions of the twisted and non-twisted bilayer BP display isotropic and anisotropic electronic dispersions, respectively. Additionally, the singly degenerate conduction band minima (CBM) and valence band maxima (VBM) states in non-twisted bilayer BP are a consequence of the

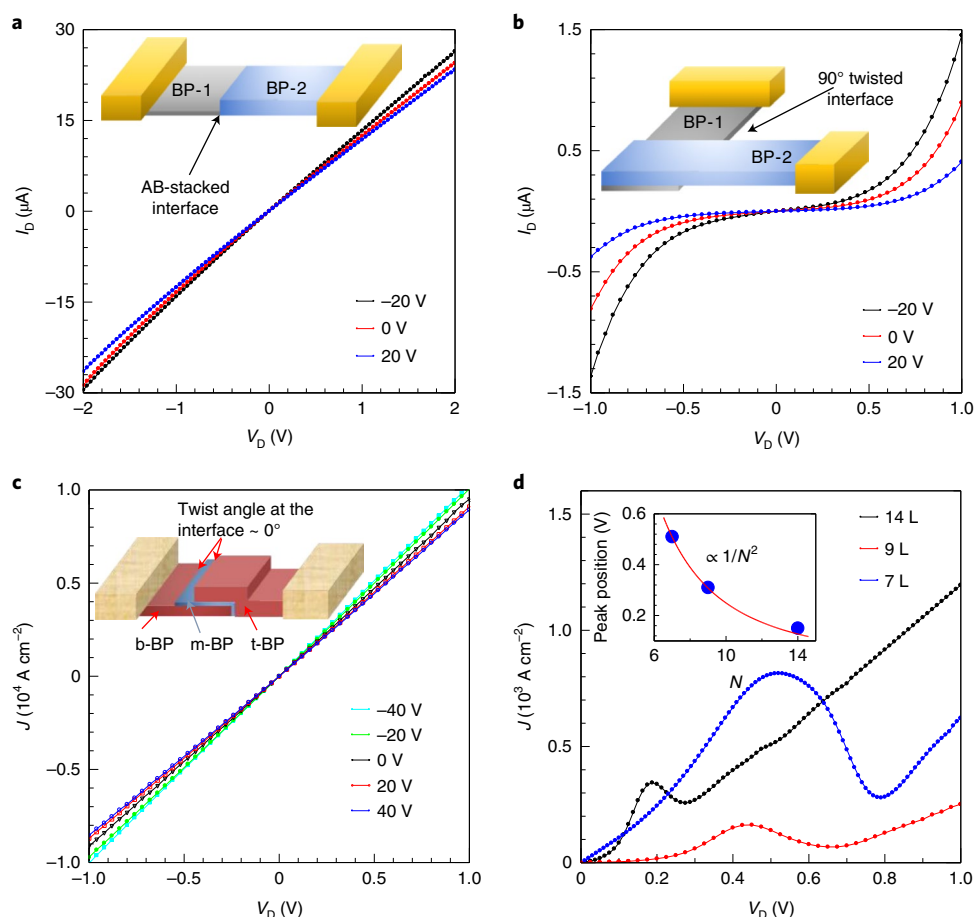


Fig. 2 | Current transport across BP bilayer and trilayer homojunctions. **a,b**, Two-terminal current–voltage characteristics of linearly stacked BP bilayers BP-1 (9 nm) and BP-2 (58 nm) (**a**) and orthogonally stacked BP bilayers BP-1 (12 nm) and BP-2 (72 nm) (**b**). Schematics of the measured devices are provided in the insets. **c,d**, Current–voltage characteristics of the trilayer linear homojunction with t-BP and b-BP thicknesses of >50 L and m-BP thickness of 7 L (a schematic of the device is provided in the inset) (**c**) and the trilayer orthogonal homojunction devices with m-BP (7–14 L) 90° twisted compared to t-BP (>50 L) and b-BP (>50 L) (**d**). The inset in **d** shows the variation in NDR peak position with number of layers N in the m-BP region. A variation in peak position $\propto 1/N^2$ suggests carrier tunnelling through QWSs formed in the m-BP region.

strong interlayer interaction, which splits the previously degenerated CBM (VBM) states of decoupled bilayer BP²⁸. However, the symmetries of the VBM states of BP monolayers give rise to negligible interlayer couplings in the bilayer structure at the twist angle of 90° . Subsequently, hole states from the two layers disperse independently, as also revealed by the localization of charge density on the individual layers²⁹. Moreover, the relaxed interlayer separation of the AB-stacked bilayer BP is 3.1 Å, which is increased by ~ 0.2 Å in 90° twisted bilayer BP. This is consistent with previous theoretical reports, and the interlayer distance has similar values among the different vdW corrections available (Supplementary Table 2). Although the interlayer distance is not the only parameter that can affect interlayer coupling, it is highly probable that the increased interlayer distance results in a weak interlayer interaction and vice versa. Indeed, a recent study³⁰ has shown weaker interlayer coupling in the twisted bilayer BP. It is worth emphasizing that, in real devices composed of mechanically stacked orthogonal BP structures, the actual interlayer distance would most probably be larger than that predicted in first-principles calculations, which provide the lower-bound estimate of interlayer distance.

It is worth mentioning that the PBE generalized gradient approximation (GGA)³¹ calculation is known to underestimate the bandgap of semiconductors as a result of the self-interaction error^{32,33}. However, this does not affect the main findings of our work because

the energy bandgap in BP has a relatively weak thickness dependence when the number of layers is greater than 5–6 L (ref. ²⁸). We stress that the BP thickness in our devices is always >6 L for the narrower m-BP region, indicating that the energy bandgap (~ 0.39 eV) is similar across the t-BP/m-BP/b-BP layers. Moreover, while modelling the current–voltage characteristics in trilayer orthogonal homojunction devices, we utilized a tight-binding model obtained from the Wannierization of a self-consistent density functional theory–Green’s function (DFT–GW) calculation of multilayer BP, instead of the PBE functional (Methods and Supplementary Section 3). This model predicts an energy bandgap very close to the experimental values²⁸.

Transport measurements on BP homojunctions

Two-terminal current transport measurements were performed on both the twisted and non-twisted BP stacked devices. The drain current–voltage (I_D – V_D) characteristics of the BP bilayer homostructure with zero twist angle show ohmic behaviour at room temperature, similar to what one would expect for pristine BP of similar thickness (Fig. 2a and Supplementary Section 4). The negligible gate dependence is due to the residual doping and the thickness, which is larger than the electrostatic screening length³⁴. However, the counterpart orthogonally twisted BP bilayer device exhibits non-linear I_D – V_D characteristics. This non-ohmic transport behaviour

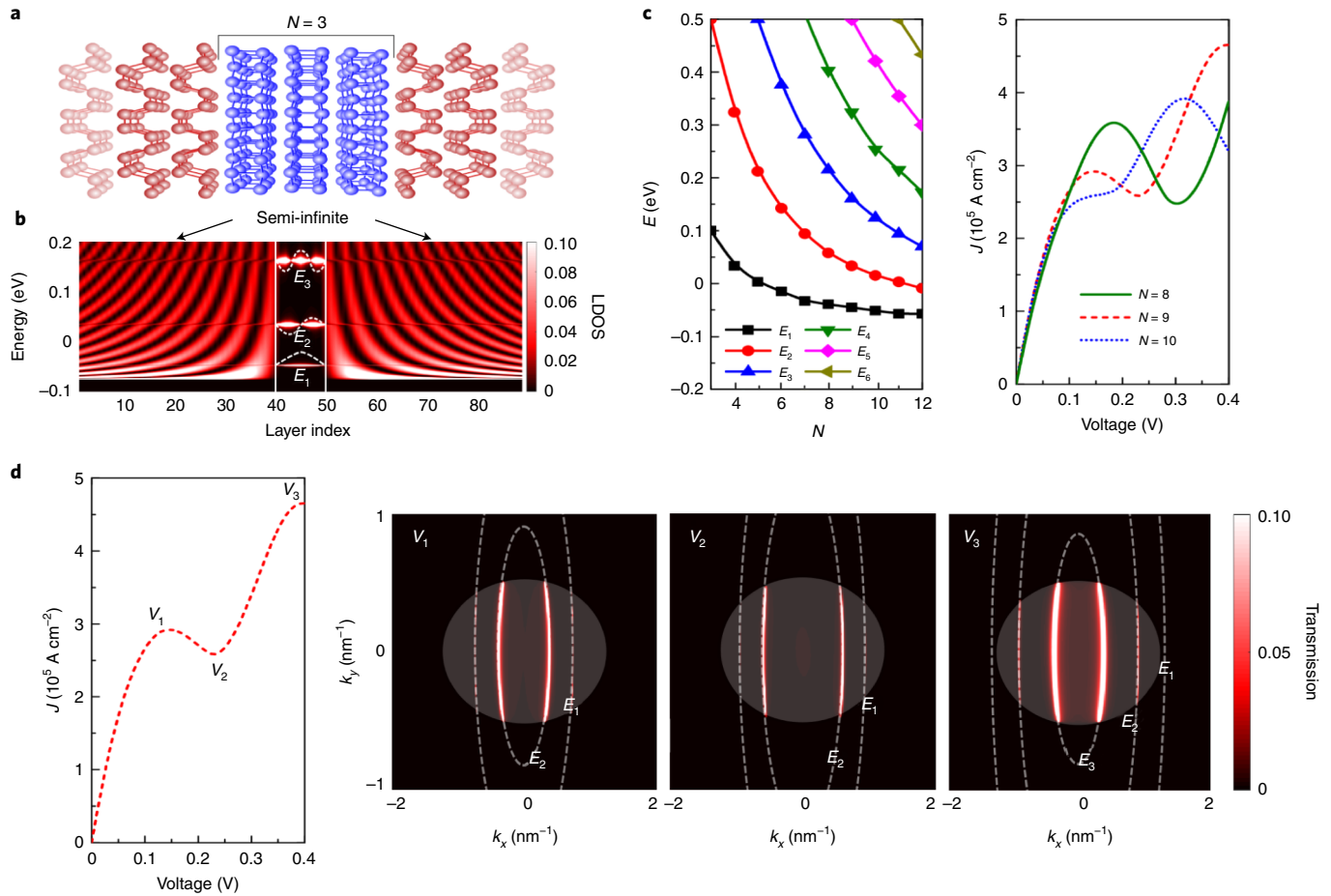


Fig. 3 | Quantum-well formation and resonant tunnelling in the trilayer orthogonal homojunction. **a**, Sketch of a BP-based trilayer orthogonal homojunction, where the middle region (represented by blue atoms) is twisted 90° in relation to the rest of the device (represented by red atoms). N is the number of layers in the twisted middle stack. Hence, in **a**, $N = 3$. **b**, Local density of states (LDOS) as a function of energy and layer index for a device with $N = 9$ in the middle region. The interface between twisted and non-twisted regions is highlighted by vertical lines. Owing to the weak interlayer interaction at the twisted interfaces and quantum confinement, bound states appear in the thin m-BP region. The energies of the first three quantized energy states are E_1 , E_2 and E_3 , respectively. **c**, Dependency of the QWS energies on the number of BP layers in the m-BP region (left) and current density as a function of applied voltage for different devices (right). **d**, The intensity maps (right) show the numerically calculated k -resolved transmission probability through a trilayer orthogonal homojunction device at different applied voltages. The selected voltages correspond to the first current peak (V_1), first current valley (V_2) and second current peak (V_3), as indicated in the current density versus voltage plot (left). The shaded ellipses in the intensity maps highlight the Fermi surface of the leads (t-BP and b-BP), while the dashed ellipse contours show the available states inside the m-BP region.

is attributed to the much weaker interlayer interaction between the twisted BP layers, and the transport is much akin to carrier tunnelling across an ultrathin vacuum layer, as is further elaborated in the following.

Next, we performed two-terminal measurements on trilayer orthogonal homojunction devices (as illustrated in Fig. 1a) with m-BP thickness varying between 4.5 nm and 7.8 nm. The respective thicknesses of the BP flakes in trilayer orthogonal homojunction devices are provided in Supplementary Section 5. Electronic transport in these devices differs distinctively, as they all exhibit NDR behaviour with maximum $\text{PVCr} \approx 3$ at $T = 300$ K (Fig. 2d). Moreover, the NDR peaks in the I_D - V_D characteristics shift to higher bias with decreasing m-BP thickness. The appearance of NDR and the evolution of peak position with m-BP thickness, as observed in trilayer orthogonal homojunction devices, suggest signatures of resonant tunnelling of carriers through bound states in the m-BP^{35,36}. On the other hand, our control device where m-BP has almost zero twist angle was found to exhibit typical linear I_D - V_D characteristics (Fig. 2c).

Quantum transport simulation

To elucidate the physics of our experimental findings in more detail, we performed quantum transport simulations based on the non-equilibrium Green's function approach (Methods and Supplementary Section 3). The electronic structure of multilayer BP is described within the tight-binding model^{28,37–39}. In this approach, a multilayer BP system is described as a series of BP layers connected to adjacent neighbours via interlayer interactions. The in-plane dynamics of low-energy electrons and holes within each monolayer, as well as the nearest-neighbour interlayer couplings, are well described by two-band block Hamiltonians, $H_0(\mathbf{k}_\parallel)$ and $H_c(\mathbf{k}_\parallel)$, respectively. The in-plane momentum $\mathbf{k}_\parallel$ is written in the general form as $\mathbf{k}_\parallel(\theta) = (k_x \cos \theta - k_y \sin \theta)\mathbf{x} + (k_x \sin \theta + k_y \cos \theta)\mathbf{y}$, where θ is the twist angle between the BP layers.

Here we simulate the trilayer orthogonal homojunction structure by considering the system sketched in Fig. 3a. The device consists of N layers of BP with $\theta = 90^\circ$ in the middle region, that is, m-BP (represented by blue atoms) connected to semi-infinite BP leads (t-BP and b-BP) with $\theta = 0^\circ$ (represented by red atoms). The interaction between

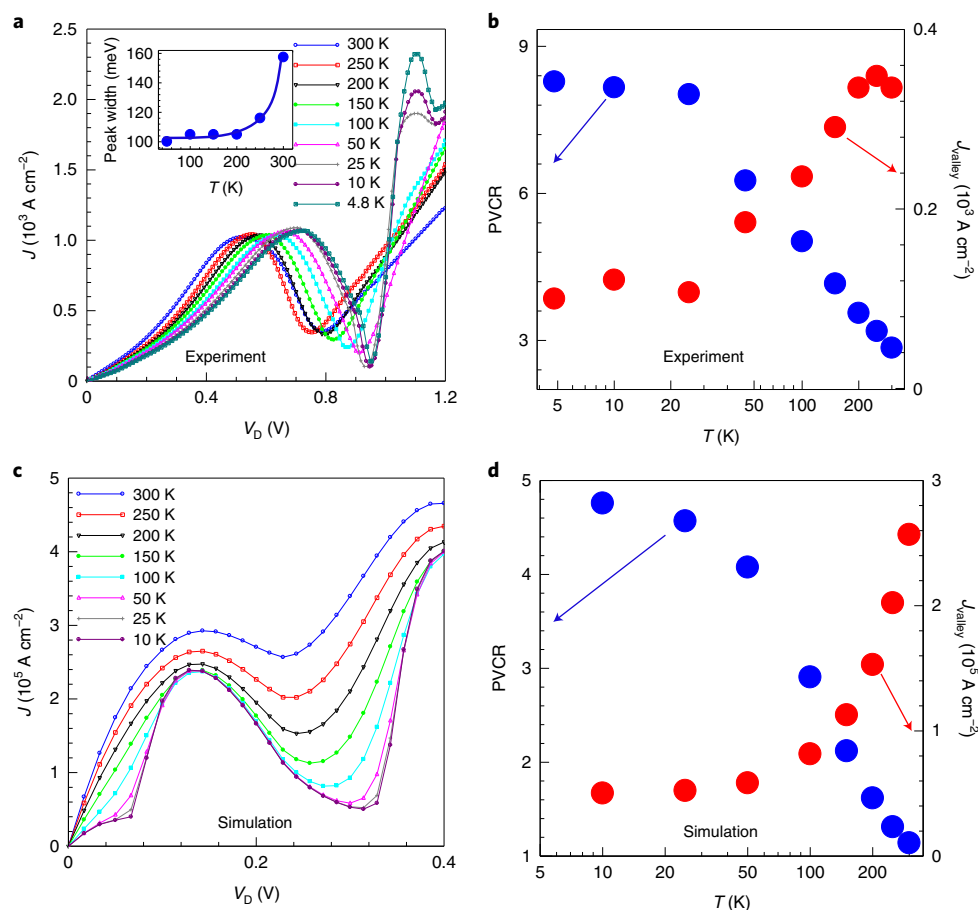


Fig. 4 | Temperature-dependent resonant tunnelling in a BP homojunction device. **a–d**, Experimental (**a,b**) and simulation (**c,d**) results for temperature-dependent current-voltage measurements on a trilayer orthogonal homojunction device with m-BP thickness of ~ 7 L. The inset in **a** shows the variation in width of the first NDR peaks with temperature. In **b** and **d**, the temperature-dependent evolution of the PVCR and valley current density (J_{valley}) of the NDR characteristics are well reproduced via quantum transport simulations of BP-based trilayer orthogonal homojunction devices.

adjacent layers within each disconnected region is well described by the AB-stacking interlayer coupling $H_c(k_{\parallel}(\theta))$ with the appropriate choice of θ . However, the interaction connecting twisted regions, $H_{\text{twisted}}(\mathbf{k}_{\parallel})$, is more challenging due to the complicated arrangement of phosphorus atoms and the larger unit cell^{29,40}. Nonetheless, first-principles calculations reveal a weaker and isotropic interlayer interaction in 90° twisted bilayer BP (Fig. 1e and refs. 29,40). Thus, we parametrize $H_{\text{twisted}}(\mathbf{k}_{\parallel}) = \alpha[H_c(\mathbf{k}_{\parallel}(0^\circ)) + H_c(\mathbf{k}_{\parallel}(90^\circ))]$, where α is an adjustable numerical parameter smaller than $\frac{1}{2}$, to ensure a twisted interaction weaker than the regular AB-stacking interlayer interaction.

We display in Fig. 3b the local density of states (LDOS) at the zone centre, for multilayer m-BP with $N=9$ twisted layers sandwiched between two semi-infinite leads. The two solid vertical white lines highlight the interface between the twisted regions. The results clearly show the existence of quantized states in the m-BP layers, where we denote the discrete energy levels as E_i (with $i=1, 2, 3, \dots$). It is worth emphasizing that these QWSs originate from the confinement of the electronic wavefunction due to the much weaker interlayer interaction between twisted m-BP and t-BP and b-BP, as we explained in Fig. 1. However, unlike conventional RTDs, which rely on physical barriers, the physics here is controlled by the twist angle between the identical materials. Propagating modes injected from the semi-infinite leads are partially reflected back and transmitted due to the discontinuity at the twisted interface.

The interference between the reflected and incident electron wavefunction causes Friedel oscillations of the electronic density inside the semi-infinite leads, as shown in Fig. 3b. Additionally, multiple reflection of the partially transmitted states take place inside the m-BP, leading to quantized confined states with standing-wave nodal patterns (Fig. 3b).

In the left panel of Fig. 3c, we show the evolution of the quantized energy states as a function of the number of layers N in the m-BP region at zero applied voltage. We found that the best fit goes as $1/N^2$ for all levels, corroborating a quantum-well-like nature of these quantized states. Hence, one anticipates a resonant tunnelling effect, similar to double-barrier RTDs, where the number of resonant peaks within a given voltage range is determined by N . The right panel of Fig. 3c shows the voltage dependence of the current density considering trilayer orthogonal homojunction devices with a varying number of layers in the m-BP. In these results, we have assumed the applied voltage to drop linearly across the twisted m-BP, with $\alpha=0.05$ and $T=300$ K. This assumption is based on the consideration that the semi-infinite leads are degenerately doped (Supplementary Sections 1 and 3). We observe NDR for all cases, in agreement with our experimental measurements. In addition, the results indicate that the peak voltage decreases with N . This is attributed to the evolution of the quantized energy states with m-BP thickness.

A detailed analysis of the tunnelling mechanism is presented in Fig. 3d, where we display the voltage-dependent current density for

the device with $N=9$ and the k -resolved transmission probability at the Fermi level at three different applied voltages denoted V_1 , V_2 and V_3 , as indicated in Fig. 3d. The shaded ellipses show the Fermi surface of the t-BP/b-BP, whereas the dashed ellipse contours show the k -resolved quantized states available in the m-BP. Resonant tunnelling takes place for those injected modes with energy and momenta matching with those of the QWSs due to energy and momentum conservation. In the absence of any mode-matching, direct tunnelling from left to right leads (that is, t-BP to b-BP) can take place, but with a much smaller transmission probability. With applied voltage V_1 , where the first current peak occurs, there is a finite region in the in-plane Brillouin zone at the vicinity of the zone centre with available QWSs with energy and momenta matching those of the leads, as shown by the high-intensity transmission. We observed that the ‘valley’ in the current–voltage characteristics occurs around V_2 , when the available QWSs with energy and momentum matching those of the leads have larger in-plane momenta. Qualitatively, the transmission probability through a barrier (in our case, a twisted interface) is highest for normally incident modes and falls rapidly as the incidence angles (or in-plane momenta) increase. This can be understood by considering free-electron models, where the transmission probability is proportional to $e^{-\kappa d}$, where $\kappa = \sqrt{k_{\parallel}^2 + k(E_F)^2}$ and $k(E_F)$ is some function of the Fermi energy.

For modes with non-vanishing in-plane momenta, that is, $k_{\parallel} \neq 0$, the effective penetration length, $1/\kappa$, is smaller, leading to much smaller transmission probabilities. At voltage V_3 , the third QWS becomes available, with modes matching those of the injected states closer to the zone centre, leading to an additional increase in tunnelling current, and hence completes the NDR characteristics. The simulation results clearly show that the peak and valley currents in the trilayer orthogonal homojunction depend on the mode-matching condition between available momentum states at the Fermi energy in t-BP/b-BP and the QWS in m-BP.

Because the NDR effect occurs due to the weak interlayer coupling between adjacent layers, one can anticipate that the twist angle variation will influence the NDR phenomenon. To address this conjecture, we fabricated additional trilayer BP homojunction devices with m-BP twisted at $\theta=30^\circ$, 45° and 60° in relation to t-BP and b-BP (Supplementary Fig. 5). Devices with m-BP twisted at $\theta=30^\circ$ and 60° exhibit NDR characteristics, that is, RTD behaviour similar to the trilayer orthogonal BP homostructures. By contrast, akin to the trilayer linear BP homojunction ($\theta \approx 0^\circ$), ohmic current transport is observed in devices with m-BP twisted at $\theta=45^\circ$. Such non-monotonic twist angle-dependent interlayer resonant tunnelling is non-trivial and could be explained by periodic variation in the interlayer coupling energy of BP bilayers with varying twist angles³⁰ (Supplementary Section 6). This result further corroborates the underlying mechanism for the NDR effect whereby the interlayer interaction determines the tunnelling between adjacent layers.

Temperature-dependent resonant tunnelling behaviour

In addition, one could expect that the quality factor of the resonant feature will be most well resolved if the thermal smearing of electrons is sufficiently smaller than the QWS energy difference. Temperature-dependent I_D – V_D measurements were carried out down to 4.8 K on trilayer orthogonal homojunction devices with m-BP thickness of 7 L and t-BP/b-BP thicknesses of >50 nm (Fig. 4a). Three observations can be made from Fig. 4a: (1) the peak (valley) current increases (decreases) with decreasing temperature, (2) the resonance peak shifts to higher voltage at reduced temperatures and (3) the second resonance peak starts to appear at $T \leq 25$ K. As a consequence of the decrease in valley current, the PVCR increases to 9 at $T=4.8$ K (Fig. 4b). Such behaviour could be explained by considering the temperature dependence of the Fermi–Dirac distribution of injected electronic states in equilibrium with the reservoirs (t-BP or

b-BP). Figure 4c shows the simulated temperature-dependent I_D – V_D characteristics for a system composed of $N=9$ twisted BP layers. For example, at voltage V_2 , where we observe the first current valley, the smearing of the distribution function due to thermal broadening⁴¹ increases the occupation of states closer to the resonance energies, for example, E_1 and E_2 (Fig. 3d). Therefore, the valley current is expected to increase. At voltage V_1 , where we observe the first current peak, the smearing of the distribution function decreases the occupancy of states at the resonance energy, which leads to a small drop in peak current at higher temperatures. The combined effect results in a PVCR that increases with decreasing temperature (Fig. 4d).

A discrepancy between experiment and theory is apparent in the temperature-dependent current–voltage characteristics. The robust NDR behaviour versus varying α shows the exponential decay of current density with varying twisted interlayer interaction (Supplementary Fig. 6). Hence, because of our choice of α in the transport simulations, we expect a quantitative mismatch between the experimental and simulated current densities. Furthermore, the shift of peak voltage with temperature is most probably due to the temperature-dependent contact/sheet resistance of the non-overlapped region in the trilayer orthogonal homojunction device, or the phonon contribution, both of which are not included in the simulation. The difference in the voltage, ΔV , between peak and valley current is ~ 40 meV and the PVCR is ~ 1.2 ($T=4.8$ K) for the second resonance peak that appears below 25 K. Phonon scattering would serve to alleviate the mode-matching requirement between the leads’ states and QWSs, as it offers additional pathways to satisfy energy–momentum conservation. As elucidated in Fig. 3, the mode-matching constraint plays an important role in the peak and valley features in the NDR effect. Hence, relaxing this constraint through phonon excitations would, on the one hand, increase the valley current^{42–44}, but, on the other hand, results in diminishing peak–valley feature (PVCR $\rightarrow 1$) at $T \geq 50$ K. Moreover, as shown in the inset of Fig. 4a, the resonance peak width (that is, the quality factor) decreases with decreasing temperature, which is another validation of this effect^{44,45}.

Conclusions

We have reported interlayer resonant tunnelling in a twist-controlled vdW homojunction that consists of a trilayer structure of anisotropic BP. An orthogonally twisted thin BP multilayer is sandwiched between two degenerate BP terminals, leading to weaker interlayer coupling and electronic decoupling between the twisted layers. This results in well-resolved QWSs in the thin BP multilayer, through which resonant tunnelling occurs. Unlike RTD heterostructures, which use a barrier layer, here the twist angle controls the interlayer coupling strength and the physics of the NDR effect, and the absence of a physical barrier allows higher current densities and improved PVCRs to be achieved. Further experimental and theoretical work may lead to novel transport phenomena in twisted vdW junctions with 2D anisotropic materials.

Methods

Fabrication of the BP homostructures and measurement. BP flakes were exfoliated on 300-nm-thick SiO_2/Si substrates using the mechanical exfoliation method. Individual BP flakes were then transferred on top of each other using the standard PDMS stamping technique. Specifically, to fabricate the trilayer orthogonal homojunction devices after exfoliation of BP flakes on the SiO_2 substrate, a thin BP flake was chosen and picked by a PDMS film. Next, we rotated the PDMS film by 90° and stacked onto a relatively thick BP flake supported on the SiO_2 substrate (a similar procedure was followed for 30° , 45° and 60° rotations). A thick BP flake was again stacked in the direction almost parallel to the bottom flake. Exfoliation and stacking were performed in a glove box with $\text{H}_2\text{O}:\text{O}_2 < 1$ ppm. After fabricating the BP homostructures, electrodes were defined using electron-beam lithography followed by metal deposition using electron-beam evaporation at a base pressure of 10^{-7} mbar. During the fabrication process, we avoided long exposure of our samples in ambient to protect the BP flakes from degradation. After device fabrication, all room-temperature data

were recorded under vacuum (10^{-3} mbar) at a probe station (MStech) with a Keithley source meter (2634B). Low-temperature measurements were carried out in a closed-cycle helium refrigerator and the temperature was controlled using a Cryocon temperature controller. Current transport was monitored using a Stanford Research Systems lock-in amplifier and a Keithley-2400 source meter.

Atomic force microscopy/Kelvin probe force microscopy measurements. Atomic force microscopy/Kelvin probe force microscopy (KPFM) measurements were performed with an SPI 4000 microscope (SII Nano Technology) in non-contact mode. To obtain the work-function values of the BP flakes, a KPFM tip (Pt/Ir-coated Si tip) was calibrated on a highly oriented pyrolytic graphite flake and the contact potential difference values between the tip and individual flakes were measured.

Electron microscopy to probe the interface morphology of BP homostructures. In the first step, trilayer orthogonal homostructures and trilayer linear homostructures were fabricated on SiO_2/Si substrates using a standard procedure, as already described. Focused ion beam milling was used to isolate a small portion of the homostructure, which was then transferred to a TEM grid for cross-sectional HRTEM/STEM imaging. The acceleration voltage for TEM measurements was 200 keV.

First-principles calculation of band structures. The electronic structure calculation was performed using the Vienna Ab initio Simulation Package (VASP) with projector augmented wave (PAW) pseudopotentials. The plane-wave cutoff energy was set to 400 eV and the Brillouin zone was sampled with a $15 \times 11 \times 1$, $15 \times 11 \times 4$ and $3 \times 3 \times 1$ Γ centred Monkhorst–Pack grid for the monolayer, the bulk and the bilayer BP, respectively. The vacuum spacing was 10 Å to clear out the interaction between adjacent layers. The exchange-correlation functional described by the PBE GGA was utilized. The vdW correction was obtained through Grimme's DFT-D2 method implemented in VASP. The relaxed cell guaranteed that the force on each atom was less than $0.01 \text{ eV \AA}^{-1}$ and the total energy was converged to 10^{-5} eV . The calculated lattice constants of the relaxed BP are $a = 3.32 \text{ \AA}$, $b = 4.42 \text{ \AA}$ and $c = 10.46 \text{ \AA}$, in good agreement with the experimental values ($a = 3.2 \text{ \AA}$, $b = 4.5 \text{ \AA}$ and $c = 10.4 \text{ \AA}$).

Quantum transport simulation. Coherent electronic transport is captured by the system's Green's function and broadening function $\Gamma_{L(R)} = i \left(\sum_{L(R)} - \sum_{L(R)}^+ \right)$ for left (L) and right (R) leads defined in terms of the self energies $\sum_{L(R)} = H_{S|}g_l H_{lS}$, where g_l is the surface Green's function of lead $l = L, R$ and $H_{S|}$ is the coupling matrix describing the interaction between the middle region and lead l . The transmission probability, current density and LDOS are $T(E) = \int \frac{dk_{\parallel}}{2\pi} \text{Tr}(G_L^+ G^+ G_R)$, $J = \frac{2e}{h} \int dE (f_L - f_R) T(E)$ and LDOS = $-\frac{1}{\pi} \text{ImTr}G$, respectively, where $f_{l(R)}$ is the Fermi–Dirac distribution of the left (right) reservoir (Supplementary Section 3).

We note that the tight-binding model adopted in our calculations is a well-established approach for describing the dynamics of low-energy carriers. In fact, this model has shown good agreement with experiment, even in the large-bias regime⁴⁶ for a multilayer system when screening is taken into account, and is expected to agree well for reasonable electric-field magnitudes, even when screening is not considered⁴⁷. In our transport results, we have assumed a maximum bias voltage of $\sim 0.4 \text{ V}$ across an m-BP composed of $N = 9$ monolayers. This is equivalent to an energy difference between adjacent layers of only $\sim 0.05 \text{ eV}$, which is inside the bias window where our model is accurate. Thus, for describing the qualitative coherent transport features and the NDR behaviour of the twisted homostructure, large biases beyond values our model can support were not considered.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding authors upon reasonable request.

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References

- Kluksdahl, N. C., Krizan, A. M. & Ferry, D. K. Self-consistent study of the resonant-tunnelling diode. *Phys. Rev. B* **39**, 7720–7735 (1989).
- Mendez, E. E., Wang, W. L., Ricco, B. & Esaki, L. Resonant tunnelling of holes in AlAs–GaAs–AlAs heterostructures. *Appl. Phys. Lett.* **47**, 415–417 (1985).
- Schmidt, O. G. et al. Resonant tunnelling diodes made up of stacked self-assembled Ge/Si islands. *Appl. Phys. Lett.* **77**, 4341–4343 (2000).
- Izumi, R., Sato, T., Suzuki, S. & Asada, M. Resonant-tunnelling-diode terahertz oscillator with a cylindrical cavity for high-frequency oscillation. *AIP Adv.* **9**, 085020 (2019).
- Alkeev, N. V., Averin, S. V., Dorofeev, A. A., Golant, E. I. & Pashkovskii, A. B. New terahertz mixer based on resonant-tunneling diode. In *Proc. 2007 International Kharkov Symposium Physics and Engineering of Millimeter and Sub-Millimeter Waves (MSMW)* 192–194 (IEEE, 2007).
- Alkeev, N. V., Averin, S. V., Dorofeev, A. A., Gladysheva, N. B. & Torgashin, M. Y. GaAs/AlAs resonant-tunnelling diode for subharmonic mixers. *Russ. Microelectron.* **39**, 331–339 (2010).
- Fan, S. et al. Tunable negative differential resistance in van der Waals heterostructures at room temperature by tailoring the interface. *ACS Nano* **13**, 8193–8201 (2019).
- Nourbakhsh, A., Zubair, A., Dresselhaus, M. S. & Palacios, T. Transport properties of a $\text{MoS}_2/\text{WSe}_2$ heterojunction transistor and its potential for application. *Nano Lett.* **16**, 1359–1366 (2016).
- Yan, R. et al. Esaki diodes in van der Waals heterojunctions with broken-gap energy band alignment. *Nano Lett.* **15**, 5791–5798 (2015).
- Kim, K. et al. Spin-conserving resonant tunnelling in twist-controlled WSe_2 -hBN- WSe_2 heterostructures. *Nano Lett.* **18**, 5967–5973 (2018).
- Burg, G. W. et al. Coherent interlayer tunnelling and negative differential resistance with high current density in double bilayer graphene- WSe_2 heterostructures. *Nano Lett.* **17**, 3919–3925 (2017).
- Xiong, X. et al. A transverse tunnelling field-effect transistor made from a van der Waals heterostructure. *Nat. Electron.* **3**, 106–112 (2020).
- Shim, J. et al. Phosphorene/rhenium disulfide heterojunction-based negative differential resistance device for multi-valued logic. *Nat. Commun.* **7**, 13413 (2016).
- Srivastava, P. K. et al. Multifunctional van der Waals broken-gap heterojunction. *Small* **15**, 1804885 (2019).
- Mishchenko, A. et al. Twist-controlled resonant tunnelling in graphene/boron nitride/graphene heterostructures. *Nat. Nanotechnol.* **9**, 808–813 (2014).
- Britnell, L. et al. Resonant tunnelling and negative differential conductance in graphene transistors. *Nat. Commun.* **4**, 1794 (2013).
- Fallahazad, B. et al. Gate-tunable resonant tunnelling in double bilayer graphene heterostructures. *Nano Lett.* **15**, 428–433 (2015).
- Luo, Z. et al. Anisotropic in-plane thermal conductivity observed in few-layer black phosphorus. *Nat. Commun.* **6**, 8572 (2015).
- Lee, S. et al. Anisotropic in-plane thermal conductivity of black phosphorus nanoribbons at temperatures higher than 100 K. *Nat. Commun.* **6**, 8573 (2015).
- Li, L. et al. Quantum oscillations in a two-dimensional electron gas in black phosphorus thin films. *Nat. Nanotechnol.* **10**, 608–613 (2015).
- Li, L. et al. Black phosphorus field-effect transistors. *Nat. Nanotechnol.* **9**, 372–377 (2014).
- Li, L. et al. Quantum Hall effect in black phosphorus two-dimensional electron system. *Nat. Nanotechnol.* **11**, 593–597 (2016).
- Hu, Z. X., Kong, X., Qiao, J., Normand, B. & Ji, W. Interlayer electronic hybridization leads to exceptional thickness-dependent vibrational properties in few-layer black phosphorus. *Nanoscale* **8**, 2740–2750 (2016).
- Mattevi, C. et al. Evolution of electrical, chemical and structural properties of transparent and conducting chemically derived graphene thin films. *Adv. Funct. Mater.* **19**, 2577–2583 (2009).
- Liu, X. et al. Modulation of quantum tunnelling via a vertical two-dimensional black phosphorus and molybdenum disulfide p–n junction. *ACS Nano* **11**, 9143–9150 (2017).
- Gaberle, J. & Shluger, A. L. Structure and properties of intrinsic and extrinsic defects in black phosphorus. *Nanoscale* **10**, 19536–19546 (2018).
- Kou, L., Ma, Y., Smith, S. C. & Chen, C. Anisotropic ripple deformation in phosphorene. *J. Phys. Chem. Lett.* **6**, 1509–1513 (2015).
- De Sousa, D. J. P., De Castro, L. V., Da Costa, D. R., Pereira, J. M. & Low, T. Multilayered black phosphorus: from a tight-binding to a continuum description. *Phys. Rev. B* **96**, 155427 (2017).
- Cao, T., Li, Z., Qiu, D. Y. & Louie, S. G. Gate switchable transport and optical anisotropy in 90° twisted bilayer black phosphorus. *Nano Lett.* **16**, 5542–5546 (2016).
- Liu, N., Zhang, J., Zhou, S. & Zhao, J. Tuning the electronic properties of bilayer black phosphorene with the twist angle. *J. Mater. Chem. C* **8**, 6264–6272 (2020).
- Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
- Shishkin, M. & Kresse, G. Self-consistent GW calculations for semiconductors and insulators. *Phys. Rev. B* **75**, 235102 (2007).
- Perdew, J. P. Density functional theory and the band gap problem. *Int. J. Quantum Chem.* **28**, 497–523 (1985).
- Low, T. et al. Plasmons and screening in monolayer and multilayer black phosphorus. *Phys. Rev. Lett.* **113**, 106802 (2014).
- Bhattacharyya, S. et al. Resonant tunnelling and fast switching in amorphous-carbon quantum-well structures. *Nat. Mater.* **5**, 19–22 (2006).
- Schuberth, G. et al. Resonant tunnelling of holes in $\text{Si/Si}_{1-x}\text{Ge}_x$ quantum-well structures. *Phys. Rev. B* **43**, 2280 (1991).
- Rudenko, A. N., Yuan, S. & Katsnelson, M. I. Toward a realistic description of multilayer black phosphorus: from GW approximation to large-scale tight-binding simulations. *Phys. Rev. B* **92**, 199906 (2015).
- Rudenko, A. N. & Katsnelson, M. I. Quasiparticle band structure and tight-binding model for single- and bilayer black phosphorus. *Phys. Rev. B* **89**, 201408R (2014).

39. Pereira, J. M. & Katsnelson, M. I. Landau levels of single-layer and bilayer phosphorene. *Phys. Rev. B* **92**, 075437 (2015).
40. Sevik, C., Wallbank, J. R., Gülseren, O., Peeters, F. M. & Çakır, D. Gate induced monolayer behavior in twisted bilayer black phosphorus. *2D Mater.* **4**, 035025 (2017).
41. Vdovin, E. E. et al. Phonon-assisted resonant tunnelling of electrons in graphene–boron nitride transistors. *Phys. Rev. Lett.* **116**, 186603 (2016).
42. Shen, G. D., Xu, D. X., Willander, M. & Hansson, G. V. The origin of the temperature dependence in resonant tunnelling transport. In *Proc. 1991 IEEE/Cornell Conference on Advanced Concepts in High Speed Semiconductor Devices and Circuits* 84–93 (IEEE, 1991).
43. Ridley, B. K. *Quantum Processes in Semiconductors* 4th edn (Oxford Univ. Press, 1999).
44. Choi, W. S., Lee, S. A., You, J. H., Lee, S. & Lee, H. N. Resonant tunnelling in a quantum oxide superlattice. *Nat. Commun.* **6**, 7424 (2015).
45. Nguyen, L. N. et al. Resonant tunnelling through discrete quantum states in stacked atomic-layered MoS₂. *Nano Lett.* **14**, 2381–2386 (2014).
46. Li, L. L., Partoens, B. & Peeters, F. M. Tuning the electronic properties of gated multilayer phosphorene: a self-consistent tight-binding study. *Phys. Rev. B* **97**, 155424 (2018).
47. Forte, J. D. S., de Sousa, D. J. P. & Pereira, J. M. Dirac spectrum in gated multilayer black phosphorus nanoribbons. *Physica E* **114**, 113578 (2019).

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Author contributions

B.S., P.K.S. and C.L. conceived the project. Y.H. prepared the devices with help from Y.Z. for electron-beam lithography. P.K.S. and Y.H. carried out electrical measurements. D.J.P.d.S. and T.L. carried out quantum transport simulations. Y.G. and M.J. carried out DFT calculations for twisted and non-twisted BP bilayer band structures. F.A. and W.J.Y. helped with the glove box facility and its use. P.K.S., D.J.P.d.S., S.G., J.T.T., B.S., T.L. and C.L. discussed and analysed the data. P.K.S., C.L., B.S., D.J.P.d.S. and T.L. co-wrote the manuscript. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Supplementary information

**Resonant tunnelling diodes based on
twisted black phosphorus homostructures**

In the format provided by the
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Supplementary Information

Resonant tunnelling diodes based on twisted black phosphorus homostructures

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Table S1. Comparison of NDR features (*mechanism, current density and PVCR*) in various vdW heterostructures.

S.No.	Device Structure	NDR Mechanism	J_{Peak} (A/cm ²)	PVCR	References
1	Graphene/h-BN/Graphene	Resonant Tunneling	~10 A/cm ² (6 K) ~30 A/cm ² (2 K)	~1.4 (6 K) ~2 (2 K)	Ref. ¹ Ref. ²
2	BP/Al ₂ O ₃ /BP	Formation of type III Broken gap junction	130 A/cm ²	130	Ref. ³
3	BP/ReS ₂	Broken-gap heterojunction	----	~3	Ref. ⁴
4	Rotationally aligned Bi-Gr/Bi-WSe ₂ /Bi-Graphene	Resonant tunneling	~200 A/cm ² (300 K)	4 (300 K)	Ref. ⁵
5	MoS ₂ /WSe ₂	Band-to-band tunneling	----	1.6 (300 K)	Ref. ⁶
6	Twist-Controlled WSe ₂ -hBN-WSe ₂ Heterostructures	Resonant Tunneling	~60 A/cm ² (300 K)	1.2 (300 K)	Ref. ⁷
7	BP/SnSe ₂	Esaki Diode	~2 A/cm ² (80 K)	2.8 (80 K)	Ref. ⁸
8	WSe ₂ /SnSe ₂	Band-to-band tunneling	~2 A/cm ² (300 K)	4 (300 K)	Ref. ⁹
9	BP/InSe	Band-to-band tunneling	----	----	Ref. ¹⁰
10	<i>Twisted black phosphorus homostructure</i>	<i>Resonant tunneling</i>	~1000 A/cm ² (300 K)	8.5 (4.8 K)	<i>This work</i>

Table S2. First principles calculation results for bilayer BP interlayer distances for AB stacked (d_0) and 90° twisted (d). All units are in Angstrom (Å).

*Bilayer AA stacking considered; **Molecular dynamics; ***Twist angle of 70° considered

vdW correction	d_0	d	Reference
DFT-D2	3.1	3.3	This work
DFT-D2	3.1	3.3	Ref. ¹¹
DFT-D3	3.2	3.4	Ref. ¹²
vdW-DF	3.3*	3.4	Ref. ¹³
optB88-vdW	3.1	3.3	Ref. ¹⁴
Lennard-Jones (LJ) **	3.1	3.2***	Ref. ¹⁵

S1: Raman spectroscopy, Kelvin probe force microscopy and Hall carrier density in BP flakes.

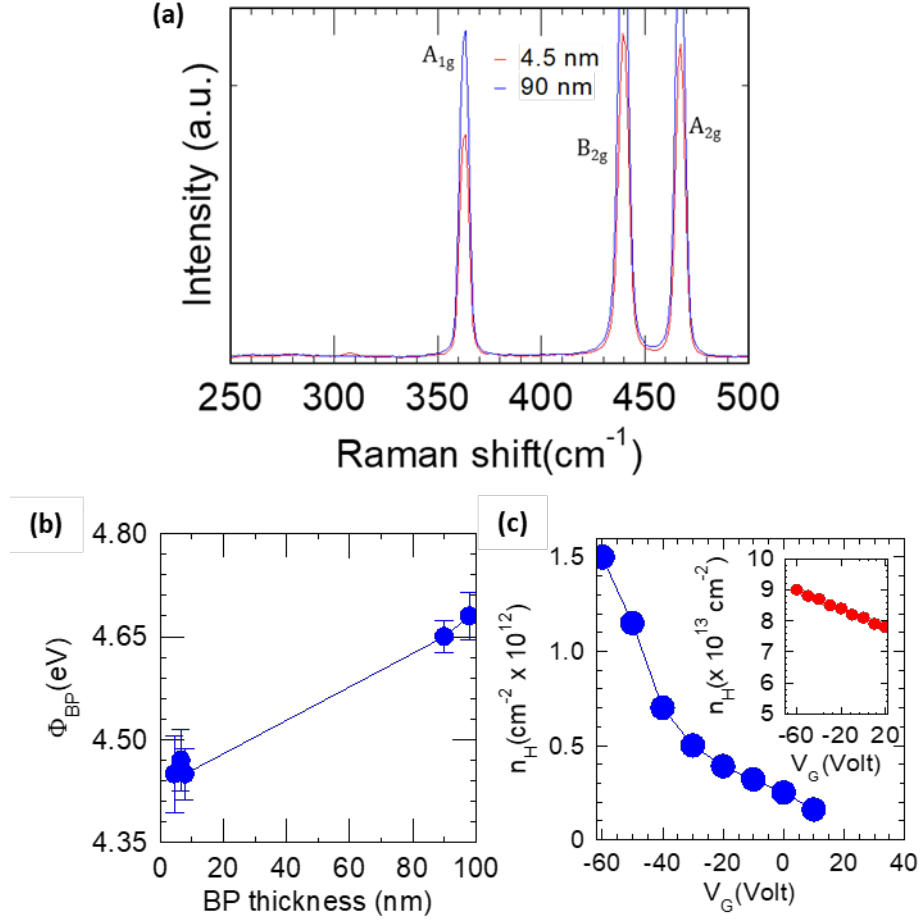


Figure S1. (a) Raman spectra of 4.5 nm and 90 nm thick BP flakes. Raman peaks for A_{1g} (365 cm⁻¹), B_{2g} (440 cm⁻¹) and A_{2g} (466 cm⁻¹) modes are labelled and supported by existing literatures.^{4,16} **(b)** Work function of BP flakes (Φ_{BP}) as a function of their respective thickness measured from AFM. Evolution of Φ_{BP} with thickness is in agreement with other reports^{17,18}. Details about Kelvin probe force microscopy (KPFM) measurement has been discussed in methods section in the main text. Depending on the Φ_{BP} values, Fermi level in BP layers > 50 nm lies deep inside the valence band maximum (4.6 eV) making the thick BP flakes a degenerate semiconductor¹⁷. **(c)** Gate dependent carrier density in 4.5 nm (main panel) and 90 nm (inset) thick BP flakes determined from Hall measurement.

S2: High resolution transmission electron microscopy (HRTEM) and scanning transmission electron microscopic (STEM) images of trilayer linear and orthogonal homojunctions.

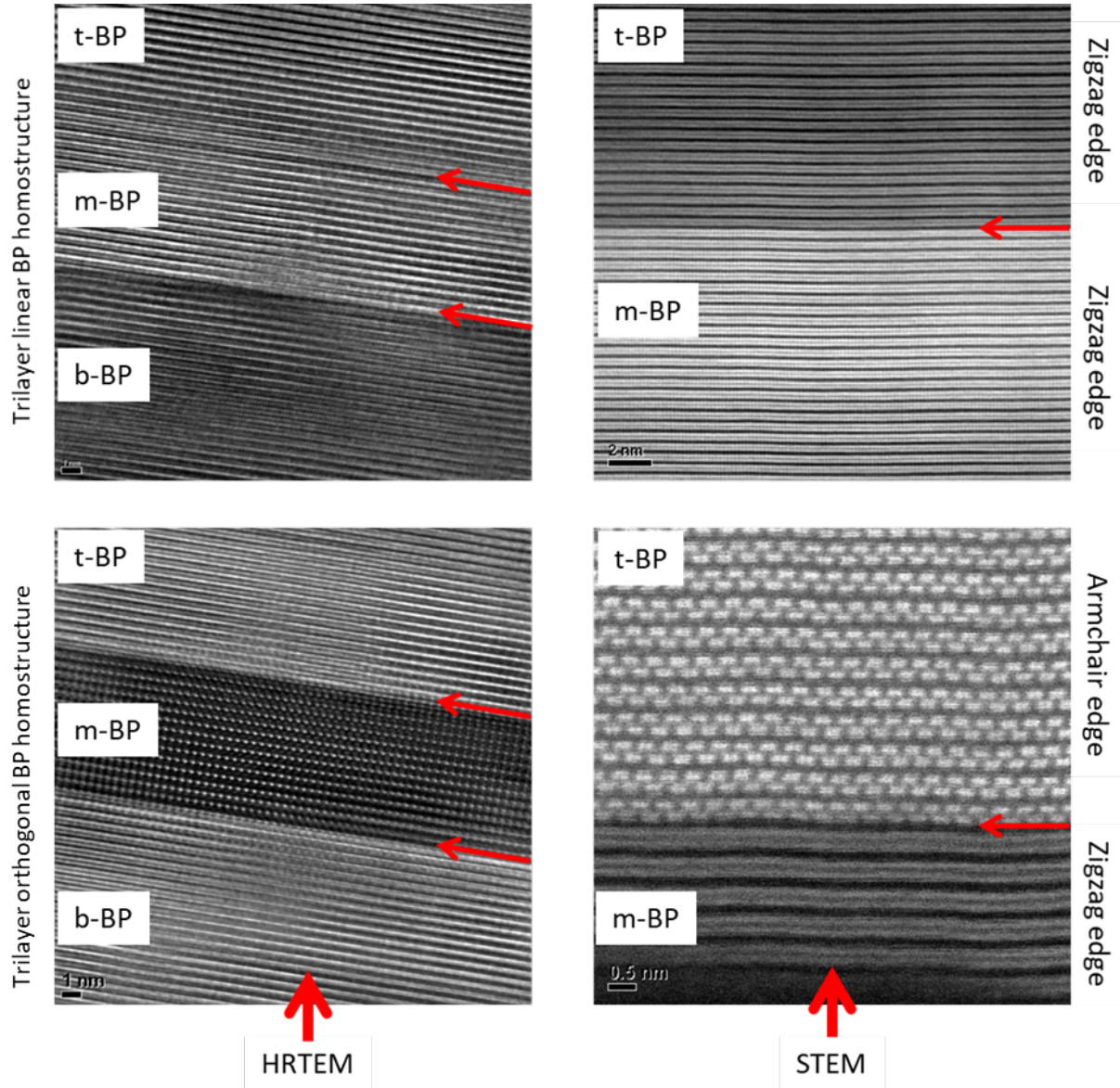


Figure S2. HRTEM and STEM images of trilayer linear homostructure (top panel) and trilayer orthogonal homostructure (bottom panel) of one of the measured devices. In both structures, m-BP thickness is around 7.8 nm. Individual flakes are labelled. STEM images were acquired near t-BP/m-BP interfaces. Appearance of armchair and zigzag directions in case of t-BP and m-BP, respectively in STEM image is a direct evidence of 90° twisted BP structures. However, the linear structure shows similar crystal directions in t-BP and m-BP. The junction between flakes are shown by arrows (These images are from different device those given in the Fig. 1 in main text).

S3. (a) Quantum Transport Simulation of BP based trilayer orthogonal homojunction devices

The tight-binding (TB) model for multilayer black phosphorus described in Ref.¹⁹ was used in conjunction with non-equilibrium Green's functions formalism to describe quantum transport in the BP based trilayer orthogonal homojunction.

The Hamiltonian for the scattering region is

$$H = \begin{pmatrix} H_L(\mathbf{k}_{||}(\theta_L)) & H_{Lm}(\mathbf{k}_{||}) & 0 \\ H_{mL}(\mathbf{k}_{||}) & H_m(\mathbf{k}_{||}(\theta_m)) & H_{Rm}(\mathbf{k}_{||}) \\ 0 & H_{mR}(\mathbf{k}_{||}) & H_R(\mathbf{k}_{||}(\theta_R)) \end{pmatrix},$$

where $H_L(\mathbf{k}_{||}(\theta_L))$, $H_m(\mathbf{k}_{||}(\theta_m))$ and $H_R(\mathbf{k}_{||}(\theta_R))$ are slab Hamiltonians for left (L), middle (m) and right (R) regions, whereas the off diagonal block-matrices describe the interlayer coupling between these regions. Here, the in-plane momenta is written in the general form $\mathbf{k}_{||}(\theta) = (k_x \cos \theta - k_y \sin \theta)\mathbf{x} + (k_x \sin \theta + k_y \cos \theta)\mathbf{y}$, where θ is the angle between the zigzag directions of the normal and rotated coordinate systems. To simulate the trilayer orthogonal homojunction, we select $\theta_L = \theta_R = 0$ and $\theta_m = 90^\circ$.

Each isolated slab Hamiltonian describes a multilayer BP system composed of N_i phosphorene layers, with $i = L, m$ or R , coupled to its adjacent neighbors and has the form

$$H_i = \begin{pmatrix} H_s(\mathbf{k}_{||}(\theta_i)) & H_c(\mathbf{k}_{||}(\theta_i)) & 0 \\ H_c(\mathbf{k}_{||}(\theta_i)) & \ddots & \ddots & \ddots & H_c(\mathbf{k}_{||}(\theta_i)) \\ 0 & H_c(\mathbf{k}_{||}(\theta_i)) & H_s(\mathbf{k}_{||}(\theta_i)) \end{pmatrix},$$

where $H_s(\mathbf{k}_{||}(\theta_i))$ is a two-band Hamiltonian describing each isolated phosphorene layer and $H_c(\mathbf{k}_{||}(\theta_i))$ is the nearest neighbor AB-stacking interlayer interaction¹⁹. We select $N_L = N_R = 1$ and $N_m = N$, i.e., we assume that middle region is composed of N 90° twisted BP layers.

The coupling between different slabs, $H_{Lm}(\mathbf{k}_{||})$ and $H_{Rm}(\mathbf{k}_{||})$, are assumed to be $H_{Lm}(\mathbf{k}_{||}) = H_{mR}(\mathbf{k}_{||}) = \alpha [H_c(\mathbf{k}_{||}(0^\circ)) + H_c(\mathbf{k}_{||}(90^\circ))]$, where α is an adjustable numerical parameter smaller than $\frac{1}{2}$, to ensure a twisted interaction weaker than the regular AB-stacking interlayer interaction. For our calculations, we have assumed $\alpha = 0.05$. We have verified that the qualitative aspects of all results remains the same for different α .

The scattering region is connected to two semi-infinite BP leads through the self-energies $\Sigma_{L(R)} = H_{Sl}g_lH_{lS}$, where g_l is the surface Green's function of lead $l = L, R$ and H_{Sl} is the coupling matrix describing the interaction between the middle region and lead l . In our model, $H_{Sl} = H_c(\mathbf{k}_{||}(\theta_l))$. The Green function of the total coupled system is

$$G(E, \mathbf{k}_{||}) = [(E + i\gamma) - H - \Sigma_L - \Sigma_R]^{-1},$$

from where one can compute the transmission probability as

$$T(E) = \int \frac{d\mathbf{k}_{||}}{(2\pi)^2} \text{Tr}(G\Gamma_L G^\dagger \Gamma_R),$$

where $\Gamma_{L(R)} = i(\Sigma_{L(R)} - \Sigma_{L(R)}^\dagger)$ are broadening functions for left (L) and right (R) leads.

(b) k-resolved local density of states

The k-resolved density of states at a given BP monolayer in the scattering region with layer index j is

$$LDOS(j, \mathbf{k}_{||}) = -\frac{1}{\pi} \text{Im}[\text{Tr}G(j, j, \mathbf{k}_{||})],$$

where $G(j, j, \mathbf{k}_{||})$ is the block matrix element of the system's green function corresponding to the j -th BP monolayer. Figure S3 shows the k-resolved LDOS for a system with $N = 9$ twisted layers. In the panel (a) we display the LDOS deep inside the lead. As one would expect, there is a continuum of available states in this case. Panels (b), (c) and (d) show the $LDOS(j, \mathbf{k}_{||})$ with $j = 4$, *i.e.*, deep inside the twisted middle region, at the three applied voltages $V1$, $V2$ and $V3$, respectively, of Figure S3 in the main text. As one can see, available states in the middle region are found only at ellipse contours corresponding to the quantum well states. Resonant tunneling will take place only for injected states with momenta matching those of the middle region, giving rise to the transmission patterns presented in the main text.

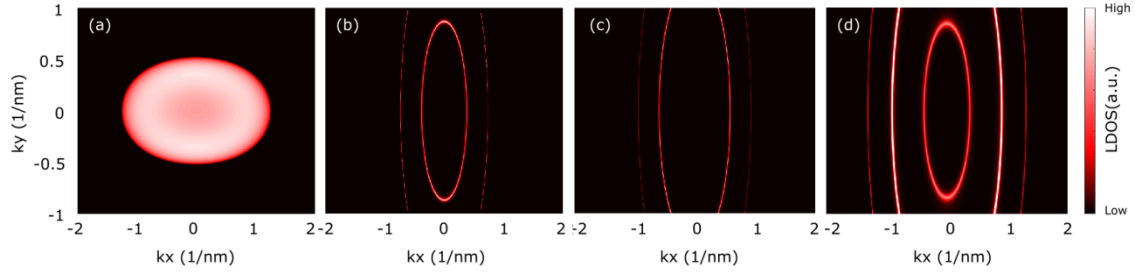


Figure S3. Layer resolved local density of states (LDOS) at the Fermi level. (a) k-resolved LDOS deep inside BP lead. In this case, one observes a continuum of available states. The k-resolved LDOS at the fourth layer inside the middle region $j = 4$ for applied voltages V_1 , V_2 and V_3 are shown in panels (b), (c) and (d), respectively. The applied voltages are defined in the main manuscript.

S4: Impact of BP flake thicknesses on the current-transport behavior in 90° twisted and non-twisted devices.

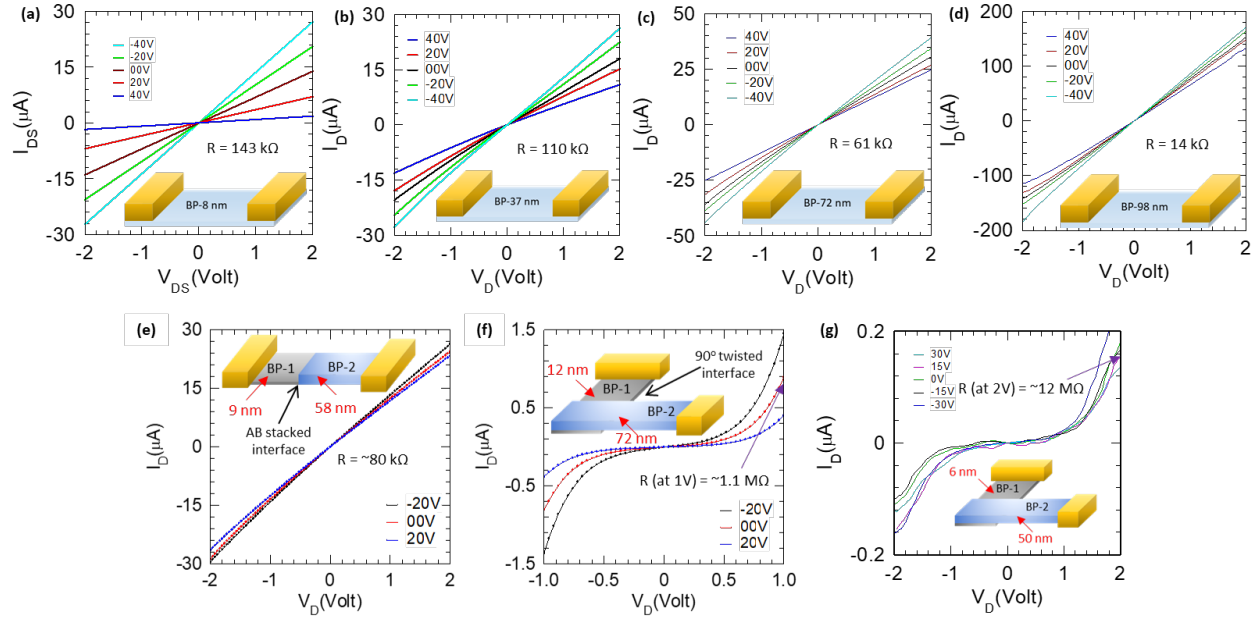


Figure. S4 (a-d) Typical current-voltage characteristics of BP FETs with varying thickness as labeled in insets. Current level (resistance) is monotonically increases (decreases) with flake thickness. Resistance values labeled in insets correspond to zero gate voltage. **(e-g)** Current-transport in (e) linearly stacked BP homostructures with constituent flake thicknesses of 9 nm and 58 nm and 90° twisted BP homostructures with constituent flake thicknesses of (f) 12 nm/72 nm and (g) 6 nm/50 nm. The data (e-f) are provided in the main text as Fig. 2a,b. Despite considerable thickness variation in devices given in (f) and (g), tunneling like current-transport can be seen.

In general, conductance of BP increases with thickness as shown in Fig. S4 (a-d) and ref.²⁰. The band structure calculation in Fig. 1f (main text) revealed strong interlayer coupling in linearly stacked BP layers. As a consequence, linearly stacked BP layers exhibit properties of a bulk BP which is manifested through Ohmic current-voltage characteristics in linearly stacked BP homojunction having accumulative thickness at the interface ~ 67 nm (Fig. S4-e). The magnitude of channel resistance in the linearly stacked BP FET (~ 80 k Ω) falls in between the channel resistances of 37 nm (110 k Ω) and 72 nm (61 k Ω) thick BP flakes. This is in agreement with the monotonic variation of conductance with BP flake thickness. Hence, the current-voltage characteristics in linearly stacked structures exhibit properties of a bulk BP and is well supported by DFT calculations.

In contrast, similar phenomenon is not true for twisted BP structures as shown in Fig. S4 (f-g). Current-transport in twisted structures is non-linear and offers high resistance $\sim 1 \text{ M}\Omega - 12 \text{ M}\Omega$. Such high resistance values are far away from channel resistance of BP FETs having similar channel thickness. Such difference in current-voltage characteristics in 90° twisted and non-twisted homojunctions was consistently observed. This ambiguity in current-voltage characteristics can be explained by considering dominant impact of momentum conservation criteria and weaker interlayer coupling at 90° twisted interface. It is worthwhile to mention that NDR feature in trilayer orthogonal BP homostructures was observed owing to the weak interlayer interaction and fulfilling the criteria of momentum conservation across the twisted interfaces.

S5: Impact of variation in t-BP and b-BP thicknesses on the resonant tunneling in trilayer orthogonal BP homojunction devices.

Sample/BP thickness (nm)	t-BP	m-BP	b-BP
Sample-1	78	4.5	55
Sample-2	65	6.5	100
Sample-3	98	7.8	56

Table S3. Summary of the thicknesses of top, middle and bottom BP flakes in trilayer orthogonal BP homojunction devices.

Clearly, t-BP/b-BP thickness in each of our trilayer orthogonal BP devices are > 50 nm and varies significantly. As evident from KPFM and Hall measurements (Section S1) BP becomes degenerate for accumulative thickness of 50 nm. For instance, we observe a substantial increase in carrier density from $2.1 \times 10^{11}/\text{cm}^2$ to $7.5 \times 10^{13}/\text{cm}^2$ when BP thickness is increased from 4.5 nm to 90 nm, respectively. In our devices, t-BP/b-BP terminals act as current injecting/collecting leads, respectively, like conventional RTDs. As shown in section S4, current-voltage characteristics in BP FETs remains linear and only magnitude of the current varies with thickness. Hence, an increase (decrease) in t-BP/b-BP thickness is supposed to affect only the tunneling current density and not the NDR peak voltage. NDR peak position in our devices solely depends on the criteria of energy and momenta matching between injected modes from emitter with those of the bound states in quantum well. Despite of the varying thicknesses of t-BP/b-BP, we have verified experimentally (manuscript Fig. 2d) that NDR peak position in our devices varies as I/N^2 where N is the number of atomic layers in m-BP. This is further reconciled with quantum transport simulations (manuscript Fig. 3c) that rules out substantial impact of t-BP/b-BP thicknesses except on the magnitude of tunneling current density. This can also be understood by looking at the energy separation of sub-bands in finite thickness t-BP and b-BP layers. The sub-band separation is $\Delta E \propto 1/L$ where L is the thickness of t-BP/b-BP. In our samples, the minimum L is at least 10 times bigger than the thickness of m-BP (Table SIII), which, together with thermal broadening of electronic distribution in the contacts, results in very small ΔE such that injected states are sufficiently close to a continuum, as described in our quantum transport calculations. Thus, the thicknesses of t-BP and b-BP are not expected to influence the formation of quantum well states in m-BP and, consequently, the NDR behavior observed in our experiments.

S6. Twist angle dependent current-transport behavior in trilayer BP homojunctions devices.

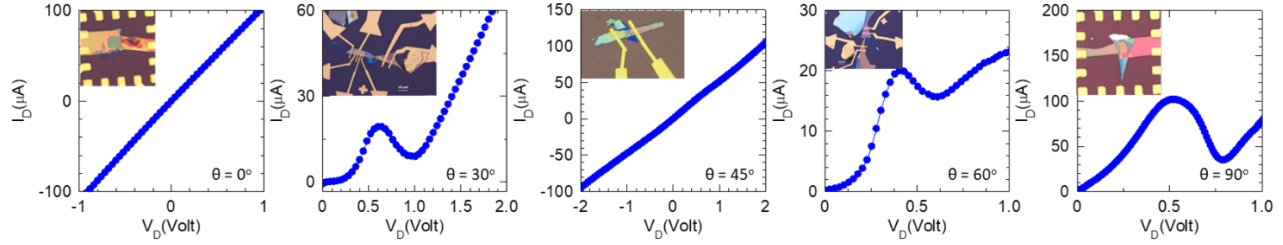


Figure S5. Current-voltage characteristics in trilayer BP homostructure devices with m-BP twisted by 0°, 30°, 45°, 60° and 90° with respect to t-BP and b-BP. Optical images of respective devices are given in insets.

Figure S5 shows the evolution in current-transport in twisted trilayer BP homojunction devices with m-BP twisted at 0°, 30°, 45°, 60° and 90° with respect to t-BP and b-BP. Measurement on these devices yield NDR characteristics in 30°, 60° and 90° twisted devices with peak-to-valley current ratio varying in the range 1.2 to 2.4. In contrast, no NDR was observed in 0° and 45° twisted devices. This non-monotonic twist angle dependent resonant tunneling could possibly be explained by considering periodic variation in interlayer coupling energy of BP bilayers with varying twist angles as proposed by *Nanshu et al.*¹². It has been proposed in ref.¹² that coupling energy between twisted BP bilayers can be approximately quantified in terms of lone pair overlaps. As a consequence, this interlayer interaction energy is non-monotonic and exhibits an oscillatory dependence on the twist angle. Owing to the fact that formation of well defined quantum well in our twisted trilayer homostructures has crucial dependence on the interlayer coupling at twisted interfaces, it is expected to observe NDR only for certain twist angles in trilayer structured devices.

S7. Effect of coupling strength (α) on the NDR characteristics in trilayer orthogonal BP homojunction devices.

We have simulated the current-voltage characteristics of a device with $N = 9$ m-BP layers at $T = 300$ K for several interlayer coupling strengths α , as shown in Fig. S6 (a): As one increases (decreases) the twisted interlayer coupling strength α , the current density can increase (decrease) by one order of magnitude. This is also apparent in Fig. S6 (b) where we show the peak current as a function of α . These results indicate that current density decays exponentially with the interlayer coupling strength. Most importantly, NDR behavior is robust against variations in α , as is apparent in Fig. S6 (a). We notice that all curves display a resonant peak, even though the magnitude of the current may vary. Additionally, the inset in Fig. S6 (b) also shows that peak-to-valley current ratio should be of the same order of magnitude for all α s, maintaining the approximate agreement between theory and experiment in Fig. 4 of the main text, but with a smaller current density.

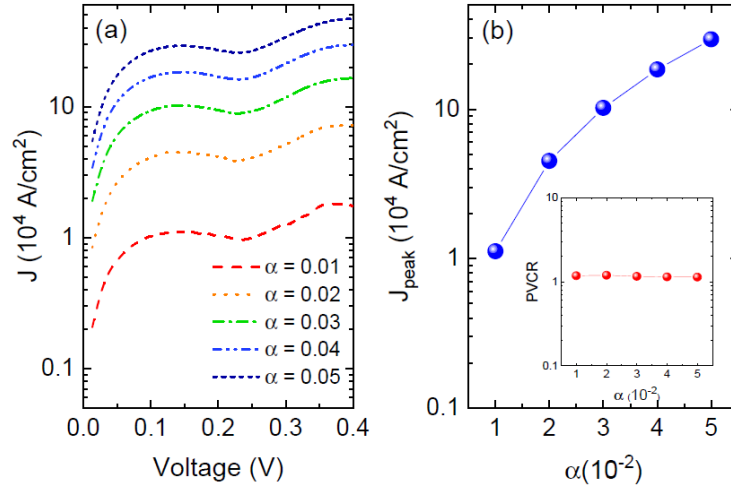


Figure S6. (a) Simulated current density as a function of applied voltage for several interlayer coupling strengths. The current density axis is plotted in log scale to emphasize the order of magnitude. (b) Dependence of peak current density with interlayer coupling strength α . The inset shows the peak-to-valley current ratio as a function of α . We have considered $T = 300$ K in all results with $N = 9$ m-BP layers.

Supplementary references.

1. Britnell, L. *et al.* Resonant tunnelling and negative differential conductance in graphene transistors. *Nat. Commun.* **4**, 1794–1795 (2013).
2. Mishchenko, A. *et al.* Twist-controlled resonant tunnelling in graphene/boron nitride/graphene heterostructures. *Nat. Nanotechnol.* **9**, 808–813 (2014).
3. Xiong, X. *et al.* A transverse tunnelling field-effect transistor made from a van der Waals heterostructure. *Nat. Electron.* **3**, 106–112 (2020).
4. Shim, J. *et al.* Phosphorene/rhenium disulfide heterojunction-based negative differential resistance device for multi-valued logic. *Nat. Commun.* **7**, 13413 (2016).
5. Burg, G. W. *et al.* Coherent Interlayer Tunneling and Negative Differential Resistance with High Current Density in Double Bilayer Graphene–WSe₂ Heterostructures. *Nano Lett.* **17**, 3919–3925 (2017).
6. Nourbakhsh, A., Zubair, A., Dresselhaus, M. S. & Palacios, T. Transport properties of a MoS₂/WSe₂ heterojunction transistor and its potential for application. *Nano Lett.* **16**, 1359–1366 (2016).
7. Kim, K. *et al.* Spin-Conserving Resonant Tunneling in Twist-Controlled WSe₂-hBN-WSe₂ Heterostructures. *Nano Lett.* **18**, 5967–5973 (2018).
8. Yan, R. *et al.* Esaki Diodes in van der Waals Heterojunctions with Broken-Gap Energy Band Alignment. *Nano Lett.* **15**, 5791–5798 (2015).
9. Fan, S. *et al.* Tunable Negative Differential Resistance in van der Waals Heterostructures at Room Temperature by Tailoring the Interface. *ACS Nano* **13**, 8193–8201 (2019).
10. Lv, Q. *et al.* Interlayer Band-to-Band Tunneling and Negative Differential Resistance in van der Waals BP/InSe Field-Effect Transistors. *Adv. Funct. Mater.* **30**, 1910713 (2020).
11. Cao, T., Li, Z., Qiu, D. Y. & Louie, S. G. Gate Switchable Transport and Optical Anisotropy in 90° Twisted Bilayer Black Phosphorus. *Nano Lett.* **16**, 5542–5546 (2016).
12. Liu, N., Zhang, J., Zhou, S. & Zhao, J. Tuning the electronic properties of bilayer black phosphorene with the twist angle. *J. Mater. Chem. C* **8**, 6264–6272 (2020).
13. Shukla, V., Grigoriev, A. & Ahuja, R. Rectifying behavior in twisted bilayer black phosphorus nanojunctions mediated through intrinsic anisotropy. *Nanoscale Adv.* **2**, 1493–1501 (2020).
14. Xie, J. *et al.* Strain-induced recovery of electronic anisotropy in 90°-twisted bilayer phosphorene. *Epl* **121**, 27002 (2018).
15. Pan, D., Wang, T. C., Xiao, W., Hu, D. & Yao, Y. Simulations of twisted bilayer orthorhombic black phosphorus. *Phys. Rev. B* **96**, 041411R (2017).

16. Srivastava, P. K. *et al.* Van der Waals Broken-Gap p–n Heterojunction Tunnel Diode Based on Black Phosphorus and Rhenium Disulfide. *ACS Applied materials & interfaces* **11**, 8266 (2019).
17. Srivastava, P. K. *et al.* Multifunctional van der Waals Broken-Gap Heterojunction. *Small* **15**, 1804885 (2019).
18. Liu, X. *et al.* Modulation of Quantum Tunneling via a Vertical Two-Dimensional Black Phosphorus and Molybdenum Disulfide *p-n* Junction. *ACS Nano* **11**, 9143–9150 (2017).
19. De Sousa, D. J. P., De Castro, L. V., Da Costa, D. R., Pereira, J. M. & Low, T. Multilayered black phosphorus: From a tight-binding to a continuum description. *Phys. Rev. B* **96**, 155427 (2017).
20. Li, L. *et al.* Black phosphorus field-effect transistors. *Nat. Nanotechnol.* **9**, 372–377 (2014).